CreateStage Fabrication

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QUOTE #CS-2026-0155

Date: March 27, 2026
Valid for: 7 days

Prepared for: CreateStage Fabrication

Job: Cantilever Sliding Gate

12 ft' wide, cantilever sliding gate, with Mild steel square tube frame, and Pickets infill. Primer and paint finish.

I am trying to build a fence, in the back of an alley, that connects the two sides of the properties fence together. The one side of the fence is 15' long, there is 12' opening, then there is a 13' fence. There will be 4 fence posts, two per each side of each fence portion. The customer wants a Cantilever gate, with the rollers on the top so there is nothing to obscure the bottom.

MATERIALS - STOCK ORDER

Material	Qty	Stock Size	Remainder	Weight	Cost
4"x4"x1/4" HSS	7	20 ft	45.5'	1709 lbs	$779.62
1/4" Plate	1	4'x8' sheet	28.8 sqft	326 lbs	$288.00
5/8" Rd Bar	68	20 ft	45.3'	1418 lbs	$1,314.67
2"x2" 11ga Sq Tube	9	24 ft	20.8'	421 lbs	$486.78
			Total Weight	*3876 lbs*	
				Material Subtotal	**$2,869.07**

CUT LIST

Piece	Material	Length	Qty	Cut Type
sq_tube_2x2_11ga - 195.2 ft	2"x2" 11ga Sq Tube	195'-2"	1	square
round_bar_0.625 - 1314.7 ft	5/8" Rd Bar	1314'-8"	1	square
hss_4x4_0.25 - 94.5 ft	4"x4"x1/4" HSS	94'-6"	1	square
plate_0.25 - 2.6 ft	1/4" Plate	2'-7.5"	1	square

DETAILED CUT LIST

Piece	Profile	Length	Qty	Cut	Notes
Gate top rail - full cantile	sq_tube_2x2_11ga	18'-0"	1	square	18 ft total gate p
Gate bottom rail - full cant	sq_tube_2x2_11ga	18'-0"	1	square	Bottom rail runs f
Gate leading vertical stile	sq_tube_2x2_11ga	9'-8"	1	square	Vertical at latch
Gate trailing vertical stile	sq_tube_2x2_11ga	9'-8"	1	square	Vertical at far en
Gate intermediate vertical s	sq_tube_2x2_11ga	9'-8"	1	square	Vertical at the 12
Gate mid-rail (horizontal st	sq_tube_2x2_11ga	11'-8"	1	square	Horizontal mid-rai
Gate mid-rail (horizontal st	sq_tube_2x2_11ga	5'-8"	1	square	Horizontal mid-rai
Gate diagonal brace - face s	sq_tube_2x2_11ga	9'-0"	2	miter_45	X-brace diagonals
Gate diagonal brace - tail s	sq_tube_2x2_11ga	6'-10"	1	miter_45	Single diagonal in
Gate face pickets - 5/8 in s	round_bar_0.625	9'-8"	35	square	4 in OC spacing ac
Gate tail pickets - 5/8 in s	round_bar_0.625	9'-8"	17	square	4 in OC spacing ac
Fence Section 1 - top rail	sq_tube_2x2_11ga	15'-0"	1	square	15 ft fence sectio
Fence Section 1 - bottom rai	sq_tube_2x2_11ga	15'-0"	1	square	15 ft fence sectio
Fence Section 1 - mid rail	sq_tube_2x2_11ga	15'-0"	1	square	Mid-height horizon

Fence Section 1 - pickets 5/	round_bar_0.625	9'-8"	45	square	4 in OC across 180
Fence Section 2 - top rail	sq_tube_2x2_11ga	13'-0"	1	square	13 ft fence sectio
Fence Section 2 - bottom rai	sq_tube_2x2_11ga	13'-0"	1	square	13 ft fence sectio
Fence Section 2 - mid rail	sq_tube_2x2_11ga	13'-0"	1	square	Mid-height horizon
Fence Section 2 - pickets 5/	round_bar_0.625	9'-8"	39	square	4 in OC across 156
Gate carriage posts (roller	hss_4x4_0.25	13'-6"	2	square	4x4 x 1/4 wall pos
Gate catch post (latch side)	hss_4x4_0.25	13'-6"	1	square	Catch post at latc
Fence Section 1 posts	hss_4x4_0.25	13'-6"	2	square	2 posts for 15 ft
Fence Section 2 posts	hss_4x4_0.25	13'-6"	2	square	2 posts for 13 ft
Post caps - flat plate welde	plate_0.25	4.5"	7	square	Seal top of each p
Roller mounting plates - wel	plate_0.25	8"	4	square	Mounting plates fo
Gate stop plate - welded to	plate_0.25	6"	1	square	Welded to catch po
Gravity latch mounting plate	plate_0.25	6"	2	square	One plate on gate
Top roller guide bracket - o	sq_tube_2x2_11ga	4'-0"	1	square	Horizontal beam sp
Guide roller bracket plates	plate_0.25	6"	2	square	Mounting plates fo

HARDWARE & PARTS

Item	Supplier	Qty	Unit	Total
Medium-duty cantilever gate roller	Estimated	1	$450.00	$450.00
Gravity gate latch (heavy-duty, for	Estimated	1	$45.00	$45.00
Gate stop/bumper (rubber or steel,	Estimated	2	$12.00	$24.00
1/2-13 x 1.5 in hex bolt (grade 5,	Estimated	16	$0.75	$12.00
1/2-13 hex nut (grade 5)	Estimated	16	$0.30	$4.80
1/2 in flat washer (USS)	Estimated	32	$0.15	$4.80
80 lb bag high-strength concrete mi	Estimated	28	$7.50	$210.00
		Hardware Subtotal		**$750.60**

Consumables

ER70S-2 TIG filler rod 1/16 in (10 lb box)	$45.00
E7018 stick electrodes 1/8 in (10 lb box)	$35.00
MIG contact tips 0.035 in	$15.00
MIG nozzle	$10.00
TIG tungsten 3/32 in (2% lanthanated)	$30.00
TIG ceramic cup #7	$15.00
Chop saw blade 14 in (metal cutting)	$24.00
6 in core drill bit (diamond, wet)	$170.00
Rags / shop towels (bundle)	$16.00
Consumable Subtotal	**$360.00**

Shop Stock (Partial Allocation)

Items your shop likely stocks - allocated proportionally to this job

ER70S-6 MIG wire 0.035 in (33 lb spool)	100% of $130.00	$130.00
75/25 Ar/CO2 shielding gas (refill)	100% of $90.00	$90.00
Argon gas (TIG, refill)	100% of $50.00	$50.00
4.5 in flap disc 60 grit	100% of $54.00	$54.00
4.5 in grinding disc	100% of $21.00	$21.00
2 in roloc disc 60 grit (die grinder)	100% of $37.50	$37.50
4.5 in wire wheel	100% of $24.00	$24.00
Rust-inhibiting primer (spray, 1 gallon)	100% of $105.00	$105.00
Black topcoat paint (spray, 1 gallon)	100% of $120.00	$120.00

Acetone (1 gallon)	100% of $15.00	$15.00
Anti-spatter spray	100% of $16.00	$16.00
Masking tape (2 in roll)	100% of $10.00	$10.00
	Shop Stock Subtotal	**$672.50**

LABOR

Process	Hours	Rate	Total
Layout & Setup	1.5	$125.00/hr	$187.50
Cut & Prep	4.0	$125.00/hr	$500.00
Fit & Tack	5.0	$125.00/hr	$625.00
Full Weld	8.0	$125.00/hr	$1,000.00
Grind & Clean	3.0	$125.00/hr	$375.00
Finish Prep	1.5	$125.00/hr	$187.50
Paint	6.0	$125.00/hr	$750.00
Hardware Install	2.0	$125.00/hr	$250.00
Site Install	22.0	$145.00/hr	$3,190.00
Final Inspection	0.5	$125.00/hr	$62.50
Total Hours	*53.5*		
		Labor Subtotal	**$7,127.50**

FINISHING

Method: Paint (in-house)

Area: 538 sq ft

Materials: $268.95

Finishing Subtotal	**$268.95**

FABRICATION SEQUENCE

Step 1: Layout and Planning

Review all drawings and dimensions. Mark out post locations on the alley concrete: 2 carriage posts (spaced to allow 18 ft gate to stack), 1 catch post at gate opening, 2 fence section 1 posts, 2 fence section 2 posts. Verify 12 ft clear opening. Confirm roller kit mounting dimensions and bolt patterns. Create a picket spacing jig from angle iron or flat bar for consistent 4 in OC placement.

Tools: tape measure, chalk line, marker, level, calculator | ~90 min

Step 2: Cut All Material - Batch Processing

Batch cut all material by profile. START with posts: 7 posts at 162 in from 4x4 HSS. Then gate frame tubes: 2 rails at 216 in, 5 verticals at 116 in, 2 mid-rails (140 in and 68 in), diagonals (measure and cut to fit later - rough cut long). Fence rails: 3 at 180 in, 3 at 156 in. PICKETS: Set chop saw stop at 116 in. Batch cut all 136 pickets (35+17+45+39). CNC plasma cut all plates: 7 post caps, 4 roller mount plates, latch plates, stop plate, guide brackets. Deburr all cuts.

Tools: chop saw, cold saw, CNC plasma table, tape measure, deburr tool | ~240 min

Step 3: Drill Roller Mount Plates

Using the cantilever roller kit as a template, drill bolt holes in all 4 roller mount plates. Verify bolt pattern matches kit. Drill guide bracket plates for guide roller hardware.

Tools: drill press, step drill, center punch, marker | ~30 min

Step 4: Build Gate Frame - Layout and Tack

Lay out gate frame on fixture table or flat floor. Position top and bottom rails (216 in) parallel, 120 in apart (outside to outside for 10 ft gate height). Place lead stile, junction stile, and tail stile vertically between rails. Square frame using diagonal measurements - CRITICAL for cantilever gate tracking. Tack all corners. Check square again. Install mid-rails (face 140 in, tail 68 in) at 60 in height, tack. Fit diagonals - measure actual diagonal distance, cut compound miters to fit, tack in place. Re-check square after every tack.

Tools: fixture table, clamps, squares, tape measure, level | ~120 min

Step 5: Weld Gate Frame - Full Welds

With frame tacked and verified square, run full welds on all frame joints. Use backstep technique on long rails to control distortion. Weld from center outward, alternating sides. MIG weld all structural frame joints - 3/16 in fillet welds. Check for square after every 4-5 welds. Weld diagonals last.

Tools: MIG welder, clamps, squares, chipping hammer, red scotch-brite pad | ~120 min

Step 6: Install Gate Face Pickets - TIG Weld

Using picket spacing jig, position pickets at 4 in OC across the gate face section (35 pickets). Start from one end, tack each picket at top rail, mid-rail, and bottom rail before moving to next. After all 35 are tacked and spacing verified, TIG weld all visible joints on the face section. Each picket gets fillet welds at 3 rail intersections = 105 weld points on face side. Grind each TIG weld smooth as you go - access between pickets is tight (3.375 in clear gap), use die grinder with 2 in roloc disc.

Tools: TIG welder, picket jig, clamps, die grinder, 2 in roloc discs | ~300 min

Step 7: Install Gate Tail Pickets - MIG Weld

Position and tack 17 pickets in tail section at 4 in OC. MIG weld all joints - tail section is hidden behind carriage posts when gate is closed. 17 pickets x 3 weld points = 51 welds. Grind cleanup pass only.

Tools: MIG welder, picket jig, clamps, angle grinder, flap disc | ~90 min

Step 8: Weld Hardware Mounting Plates to Gate

Weld latch mounting plate to gate lead stile at appropriate height for gravity latch (typically 42-48 in). Weld gate stop bumper pad location. All plates MIG welded with full penetration where structural.

Tools: MIG welder, clamps, level, tape measure | ~30 min

Step 9: Build Fence Panels - Section 1 (15 ft)

Lay out fence section 1: top, mid, and bottom rails at 180 in. If building as a panel in shop, space rails at 0 in (bottom), 60 in (mid), and 116 in (top) - pickets span between top and bottom. Tack all 45 pickets at 4 in OC. MIG weld all joints: 45 pickets x 3 rails = 135 weld points. Grind cleanup.

Tools: MIG welder, fixture table, picket jig, clamps, angle grinder | ~180 min

Step 10: Build Fence Panels - Section 2 (13 ft)

Same process as Section 1 but 156 in rails and 39 pickets. Tack all pickets, verify spacing, MIG weld all 117 weld points. Grind cleanup.

Tools: MIG welder, fixture table, picket jig, clamps, angle grinder | ~150 min

Step 11: Grind and Clean All Assemblies

Grind all weld spatter, sharp edges, and rough welds on gate and fence panels. Focus on visible gate face - TIG welds should already be clean from Step 6. Fence panels get cleanup pass only. Remove mill scale from all surfaces with flap disc or wire wheel for paint adhesion. Wipe down with acetone/solvent.

Tools: angle grinder, flap discs, wire wheel, die grinder, roloc discs | ~180 min

Step 12: Weld Post Caps and Roller Mount Plates to Posts

Weld post caps to top of all 7 posts. On the 2 carriage posts, weld roller mount plates at the correct height and position for the cantilever roller kit - the top rail of the gate must ride through the rollers at the correct elevation. Mock-fit roller assemblies to verify alignment before final welding. Weld latch plate and stop plate to catch post. Weld guide bracket plates.

Tools: MIG welder, level, clamps, tape measure, squares | ~90 min

Step 13: Prime and Paint - All Components

Apply 2 coats rust-inhibiting primer (spray) to all assemblies: gate panel, 2 fence panels, 7 posts, all hardware plates. Allow proper flash time between coats. Apply 2 coats black topcoat. Pay special attention to inside of tube ends, post bottoms (below grade section gets extra primer), and tight spaces between pickets. Mask bolt holes on roller mount plates before painting.

Tools: HVLP spray gun, compressor, masking tape, paint booth/area, respirator | ~360 min

Step 14: Site Work - Core Drill Post Holes

Core drill 6 in diameter holes through concrete at all 7 post locations, 42 in deep (Chicago frost line). Use wet core drill. Remove cores and debris. Verify hole spacing matches fabricated panels and gate.

Tools: core drill with 6 in bit, water supply, shop vac, tape measure, chalk line | ~420 min

Step 15: Set Posts in Concrete

Set all 7 posts in holes. Plumb each post with level in both directions. Brace with temporary supports. CRITICAL: Carriage posts must be perfectly plumb and at exact spacing for cantilever gate to track properly. Fill holes with high-strength concrete mix (5000 PSI). Allow minimum 24-48 hour cure before loading.

Tools: level, temporary braces, concrete mixer or bags, trowel, water | ~240 min

Step 16: Install Cantilever Roller Hardware on Posts

After concrete cure, bolt cantilever roller assemblies to the pre-drilled mounting plates on carriage posts. Install guide rollers on guide bracket plates. Verify rollers are level and aligned - gate top rail must slide freely through all rollers.

Tools: socket set, torque wrench, level, shims | ~60 min

Step 17: Hang Gate on Rollers

Lift 18 ft gate panel and feed top rail through cantilever rollers. This is a heavy lift - gate weighs approximately 800-1000 lbs. Use engine hoist, forklift, or multiple helpers. Slide gate through full range of travel. Verify: gate clears ground at all points, gate reaches full closed position against catch post stop, gate opens fully to stack behind carriage posts. Adjust roller height with shims if needed.

Tools: engine hoist or forklift, helpers (3-4 people), level, shims, pry bar | ~90 min

Step 18: Install Fence Panels

Lift fence panels into position between posts. Level and plumb. Stick weld (E7018) fence rail ends to posts in the field. Section 1 (15 ft panel)

between its 2 posts. Section 2 (13 ft panel) between its 2 posts. Verify fence top is level and aligned with gate top.

Tools: stick welder (E7018), level, clamps, helpers, ladder | ~180 min

Step 19: Install Gravity Latch and Final Hardware

Install gravity latch on gate and catch post. Adjust latch engagement. Install any gate guides, bumpers, or stops. Test gate operation through full travel - open, close, latch engagement. Verify bottom clearance is unobstructed.

Tools: socket set, drill, level, tape measure | ~45 min

Step 20: Touch-Up Paint and Final Inspection

Touch up all field welds, scratches, and handling marks with matching black paint. Inspect all welds, hardware, and operation. Verify gate rolls smoothly, latches securely, fence is plumb and level. Clean up site.

Tools: spray paint, brush, rags, level, tape measure | ~90 min

REVIEW REQUIRED

1 hardware item(s) removed - no matching fabrication step: 1/2 in lock washer

PROJECT TOTAL

Materials	$2,869.07
Hardware & Parts	$750.60
Consumables	$360.00
Shop Stock	$672.50
Labor	$7,127.50
Finishing	$268.95
Subtotal	**$12,048.62**
Markup (15%)	$1,807.29
QUOTE TOTAL	**$13,855.91**

ASSUMPTIONS

- Material prices based on market averages (Feb 2026) - update with supplier quotes for accuracy.
- Labor hours estimated by AI (claude-opus-4-6 via Claude) with domain guidance.
- Hardware prices from catalog data - verify availability and current pricing before ordering.
- Consumables estimated at $360.00 based on weld volume and finish area.
- Gate height is 10 ft (120 in) to match fence height - top of gate aligns with top of fence.
- Total cantilever gate panel is 18 ft (216 in): 12 ft face + 6 ft counterbalance tail (50% of opening).
- Posts are 4x4 HSS with 1/4 in wall, 162 in total length (120 in above grade + 42 in below grade).
- Gate frame is 2x2 square tube 11 gauge per user specification.
- Pickets are 5/8 in solid round bar at 4 in on center per user specification.
- Top-mount cantilever system: rollers mount to overhead beam/plates on carriage posts, gate top rail rides through rollers. Bottom is unobstructed.
- Two carriage posts are spaced to allow full 18 ft gate to stack when open (approximately 4 ft apart, positioned at the stacking end of the gate opening).
- One catch post at the latch end of the 12 ft opening.
- Fence panels are shop-fabricated as complete panels, then field-installed between posts with stick welding.
- Concrete ground surface - core drilling required for post holes.
- 7 total posts: 2 carriage + 1 catch + 2 fence section 1 + 2 fence section 2.
- Gravity latch is a standard off-the-shelf unit, not custom fabricated.
- Medium-duty cantilever roller kit purchased as a kit - not custom engineered.
- Primer is rust-inhibiting type suitable for outdoor mild steel. Topcoat is industrial enamel, black.
- Fence section rails run between posts - rails are full section length, posts are at each end.
- Picket count: gate face 35, gate tail 17, fence 1 = 45, fence 2 = 39. Total 136 pickets.
- Gate weight estimated 800-1000 lbs - requires mechanical lift for installation.
- Chicago frost line is 42 inches - all posts set to this depth minimum.
- Cut list generated by AI from project description.
- Hardware total is $750.60 - consider consolidating orders for volume discounts. Potential savings: 5-10%.

EXCLUSIONS

- Permit fees and engineering review
- Demolition or removal of existing work (unless explicitly included)
- Concrete work beyond post holes
- Permit acquisition - Chicago may require building/fence permits. Not included.
- Engineering stamp or structural calculations - not included. Recommend PE review for 10 ft cantilever gate.
- Utility locates (JULIE/811 call) - customer responsibility before core drilling.
- Gate operator/motor - customer specified manual slide. Add motorization separately if needed later.
- Decorative elements (scrollwork, finials, etc.) - not described, not included.
- Removal or modification of existing property fences at connection points.
- Concrete repair/patching around core-drilled post holes.
- Drainage or grading work.
- Any work above 10 ft height (no overhead structure or canopy).
- Galvanizing - finish is primer and paint per customer request.
- Spare parts or maintenance kit for roller hardware.

This quote is valid for 7 days from the date above.
Payment terms: Deposit due at signing: $5,350.00 (100% of materials) + $4,098.31 (50% of labor) = $9,448.31 total deposit. Remaining balance of $4,407.60 due upon completion.